

Crops

|              | Crop 1     | Crop 2  | Crop 3 | Crop 4  | Crop 5 | Crop 6 | Crop 7 | Crop 8     | Crop 9       | Crop 10 | Crop 11         |
|--------------|------------|---------|--------|---------|--------|--------|--------|------------|--------------|---------|-----------------|
|              | Horse Bean | Einkorn | Emmer  | Lentils | Peas   | Olives | Grapes | Grass Peas | Bitter Vetch | Barley  | FreeHull Barley |
| Yield        | 500        | 500     | 500    | 300     | 150    | 600    | 600    | 500        | 400          | 500     | 500             |
| Units        | Ha         | Ha      | Ha     | Ha      | Ha     | Ha     | Ha     | Ha         | Ha           | Ha      | Ha              |
| Mcals/kg     | 3          | 3       | 3      | 3       | 3.25   | 4      | 3      | 3          | 3            | 3       | 3               |
| Fallow Ratio | 1          | 1       | 1      | 1       | 1      | 0      | 0      | 1          | 1            | 1       | 1               |
| Fallow MCals | 500        | 500     | 500    | 500     | 500    | 500    | 500    | 500        | 500          | 500     | 500             |
| Fodder Kg    | 300        | 300     | 300    | 300     | 100    | 0      | 0      | 300        | 300          | 300     | 300             |
| Fodder MCals | 0.8        | 0.8     | 0.8    | 0.8     | 0.8    | 0.8    | 0.8    | 0.8        | 0.8          | 1.42    | 0.8             |

Animal Definitions

|                                      | Animal 1 | Animal 2 | Animal 3 | Animal 4 | Animal 5 | Animal 6 | Animal 7 | Animal 8 |
|--------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|
|                                      | Sheep    | Pig      | Cow      | Goat     |          |          |          |          |
| Mcalories Kg <sup>-1</sup>           | 3.200    | 3.200    | 3.200    | 3.200    |          |          |          |          |
| Usable %                             | 50.00%   | 50.00%   | 50.00%   | 50.00%   |          |          |          |          |
| Average Kill Weight                  | 30       | 100      | 400      | 30       |          |          |          |          |
| Weaning Weight                       | 8        | 2        | 50       | 8        |          |          |          |          |
| Adult Weight                         | 60       | 200      | 625      | 60       |          |          |          |          |
| Grow Time                            | 40       | 40       | 50       | 40       |          |          |          |          |
| Death Rate                           | 30.00%   | 30.00%   | 20.00%   | 30.00%   |          |          |          |          |
| Conception Efficiency                | 80.00%   | 70.00%   | 85.00%   | 80.00%   |          |          |          |          |
| Sex Maturity                         | 18.00    | 18       | 18       | 18       |          |          |          |          |
| Breeding Years                       | 5        | 5        | 8        | 5        |          |          |          |          |
| Babies Birth <sup>-1</sup>           | 1        | 6        | 1        | 1        |          |          |          |          |
| Wean Age                             | 60       | 30       | 60       | 60       |          |          |          |          |
| Gestation                            | 150      | 115      | 279      | 150      |          |          |          |          |
| Oestrous Cycle                       | 21       | 21       | 21       | 21       |          |          |          |          |
| Lactation Time                       | 210      | 30       | 240      | 210      |          |          |          |          |
| Ratio F:M breed stock                | 20       | 20       | 20       | 20       |          |          |          |          |
| Desired Herd Size                    | 50       | 0        | 10       | 50       |          |          |          |          |
| Mixed Herd Numbers                   | 25       | 0        | 2        | 25       |          |          |          |          |
| Milk grams day <sup>-1</sup>         | 0.94     | 0        | 6        | 0.9      |          |          |          |          |
| Milk MCals Kg <sup>-1</sup>          | 0.940    | 0        | 0.94     | 0.940    |          |          |          |          |
| Ewe Fleece Kg Yr <sup>-1</sup>       | 2.25     |          |          |          |          |          |          |          |
| Ram Fleece Kg year <sup>-1</sup>     | 1.75     |          |          |          |          |          |          |          |
| Juvenile Fleece Kg Yr <sup>-1</sup>  | 1.4      |          |          |          |          |          |          |          |
| Gestation Req'd                      | 3.970    | 3.970    | 6        | 3.970    |          |          |          |          |
| Lactation Req'd                      | 5.410    | 5.410    | 7.5      | 5.410    |          |          |          |          |
| Maintenance Req'd                    | 3.200    | 0        | 5.8      | 3.2      |          |          |          |          |
| Juvenile's Req kg <sup>-1</sup>      | 0.030    | 0.030    | 0.030    | 0.030    |          |          |          |          |
| Land Energy Hectare <sup>-1</sup>    | 1500     | 1500     | 1500     | 1500     |          |          |          |          |
| Fodder Fed From Crop                 | Crop 10  | Crop 10  | Crop 10  | Crop 10  |          |          |          |          |
|                                      | Barley   | Barley   | Barley   | Barley   |          |          |          |          |
| Grams <sup>-1</sup> Straw/Chaff      | 1000     |          | 4956     | 1000     |          |          |          |          |
| Grams <sup>-1</sup> Grain            | 750      | 750      | 1500     | 750      |          |          |          |          |
| Days on supplemental feeding program | 90       | 90       | 90       | 90       |          |          |          |          |

CROP TARGETS

|                       | Percentage | MCals for   | Calorie Tgt | Percent | Percent for Prdn Tgt | Area Target | Fallow MCals | Straw/Chaff MCal |
|-----------------------|------------|-------------|-------------|---------|----------------------|-------------|--------------|------------------|
|                       | Tame Diet  | Animal Feed | Mcal        | Wastage | re-seeding           | Kg          | Ha           | Available        |
| TRUE Horse Bean       | 1.09%      | 0           | 627         | 30%     | 20%                  | 313         | 1            | 313              |
| TRUE Einkorn          | 5.47%      | 0           | 3145        | 30%     | 20%                  | 1572        | 6            | 1572             |
| TRUE Emmer            | 52.52%     | 0           | 30192       | 30%     | 20%                  | 15096       | 60           | 15096            |
| TRUE Lentils          | 1.09%      | 0           | 627         | 30%     | 20%                  | 313         | 2            | 522              |
| TRUE Peas             | 3.28%      | 0           | 1886        | 30%     | 20%                  | 870         | 12           | 2901             |
| TRUE Olives           | 1.53%      | 0           | 880         | 30%     | 0%                   | 286         | 0            | 238              |
| FALSE Grapes          | 0.00%      | 0           |             | 30%     | 0%                   |             |              |                  |
| FALSE Grass Peas      | 0.00%      | 0           |             | 30%     | 20%                  |             |              |                  |
| TRUE Bitter Vetch     | 2.19%      | 0           | 1259        | 30%     | 20%                  | 629         | 3            | 787              |
| TRUE Barley           | 32.83%     | 49637       | 68510       | 30%     | 20%                  | 34255       | 137          | 34255            |
| FALSE FreeHull Barley | 0.00%      | 0           |             | 30%     | 20%                  |             |              | 29185            |
| TOTALS                | 100.00%    | 49637       | 107125      |         |                      | 53336       | 222          | 55685            |
|                       |            |             |             |         |                      |             |              | 29185            |

ANIMAL TARGETS

|            | Percentage | Calorie Tgt | Prdn Tgt  | Total MCals | MCals From  | MCals From | MCals From  | MCals From | Ha       |
|------------|------------|-------------|-----------|-------------|-------------|------------|-------------|------------|----------|
|            | Tame Diet  | Mcal        | Kg        | Feed Reqd   | Straw/Chaff | Grain      | Fallow Land | Land       | Required |
| TRUE Sheep | 3.59%      | 973         | 304       | 64,689      | 3,639       | 5,765      | 6,131       | 49,154     | 33       |
| TRUE Pig   | 36.65%     | 9,933       | 3,104     | 116,393     | 0           | 14,515     | 11,032      | 90,847     | 61       |
| TRUE Cow   | 47.81%     | 12,957      | 4,049     | 193,117     | 16,340      | 10,449     | 18,304      | 148,024    | 99       |
| TRUE Goat  | 11.95%     | 3,239       | 1,012     | 213,320     | 11,933      | 18,908     | 20,219      | 162,260    | 108      |
| 4 TOTALS   | 100.00%    | 27,101.3    | 8,469.141 | 587,519.2   | 31,912.4    | 49,637.1   | 55,685.1    | 450,284.7  | 300.2    |